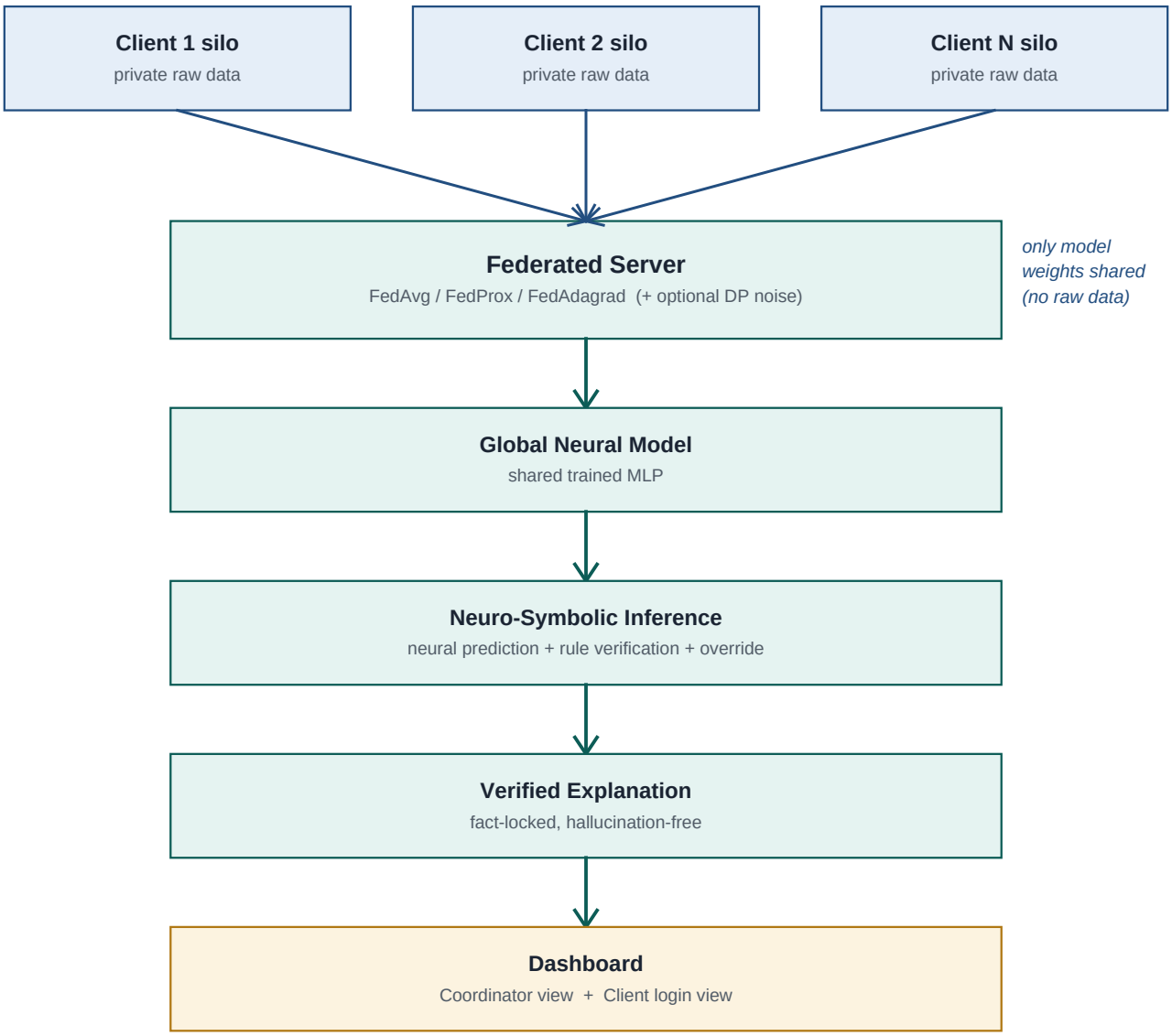
System Design

Domain-agnostic federated neuro-symbolic framework for reliable, explainable reasoning over privacy-preserving heterogeneous data (Healthcare risk and Banking fraud)

1. High-Level System Design

The big picture. Each participant (client) keeps its own private data. A federated server trains one shared neural model by exchanging ONLY model weights - never the raw data. The trained global model feeds a neuro-symbolic inference stage that verifies every prediction against domain rules, a fact-locked explainer produces a safe explanation, and results are served through a dashboard with separate coordinator and client views.

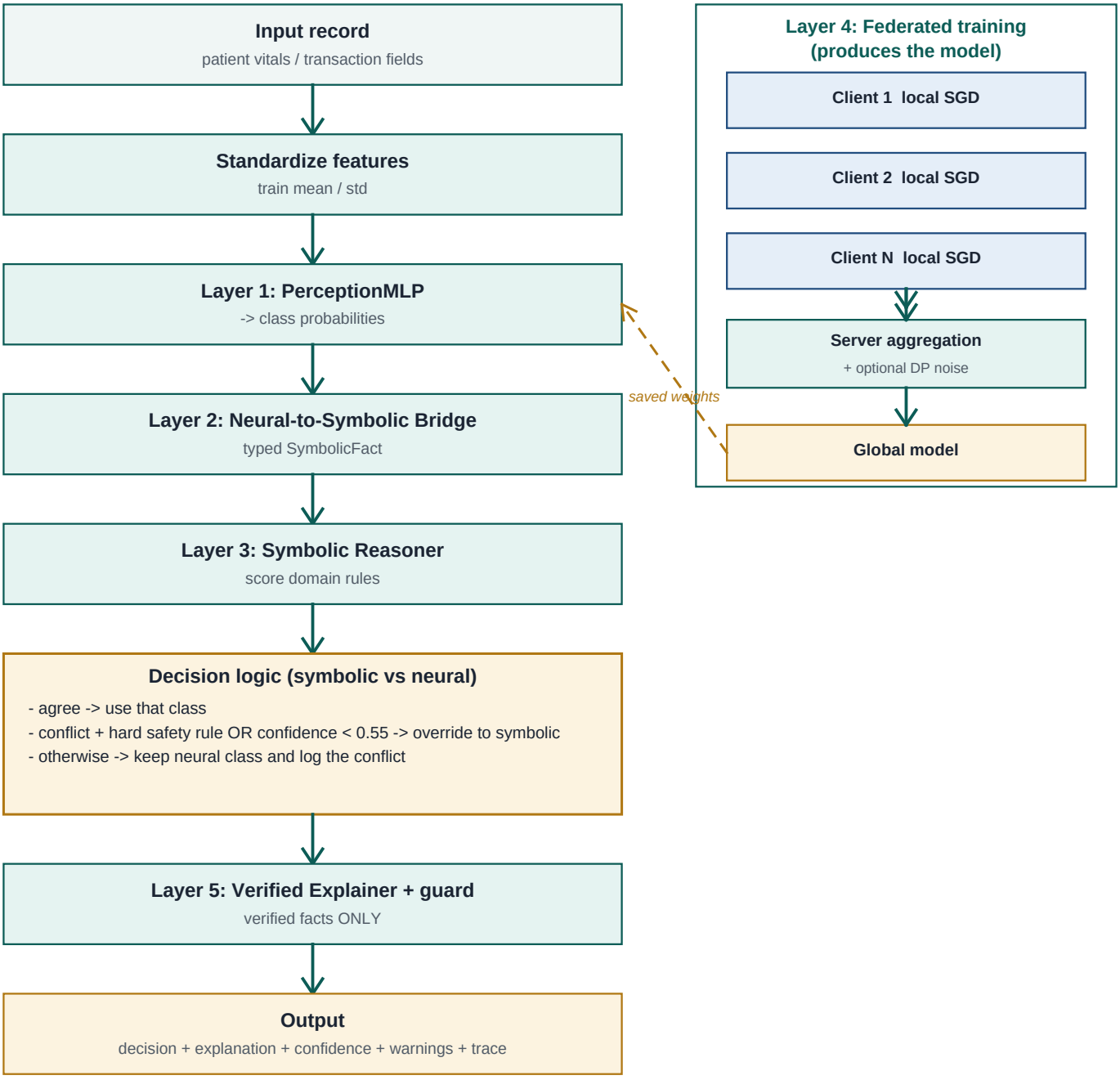


Privacy is structural: because only weights move upward, no client's records are ever centralised, yet the global model still benefits from every client's patterns.

NSF system design (high-level and detailed)

2. Detailed System Design

The internals. The left spine shows how a SINGLE record flows through the five layers at inference time, including the decision logic that resolves conflicts between the neural and symbolic layers. The right-hand panel shows Layer 4 (federated training), which runs beforehand to produce the model that Layer 1 loads.



Reading the diagram: a record is standardised, classified by the federated MLP (Layer 1), converted into a typed fact (Layer 2) and scored by the symbolic rules (Layer 3). The decision block keeps the neural class when both agree, overrides to the symbolic class when a hard safety rule fires or neural confidence is below 0.55, and otherwise keeps the neural class while logging the conflict. The verified explainer (Layer 5) then writes a fact-locked explanation, and the output bundles the decision, confidence, warnings and a full trace.

Key design decisions

- Separation of concerns: neural model for accuracy, symbolic engine for verifiable logic, explainer for safe natural language - each is independently testable and swappable.

- Privacy by construction: federated averaging plus optional differential-privacy noise on client updates; raw data never leaves a client.
- Safety overrides: hard clinical/financial constraints can override an unsafe or low-confidence neural output.
- No hallucination: the explainer sees only verified facts and passes a guard; failures fall back to a strict template, giving a 100% hallucination-free ratio in our runs.
- Domain-agnostic: the same pipeline serves healthcare and banking by swapping only the inputs and the rule set.
- Role-based access: the dashboard enforces coordinator vs client views; a logged-in client sees only its own data.